MIL-S-81551/1A(AS)

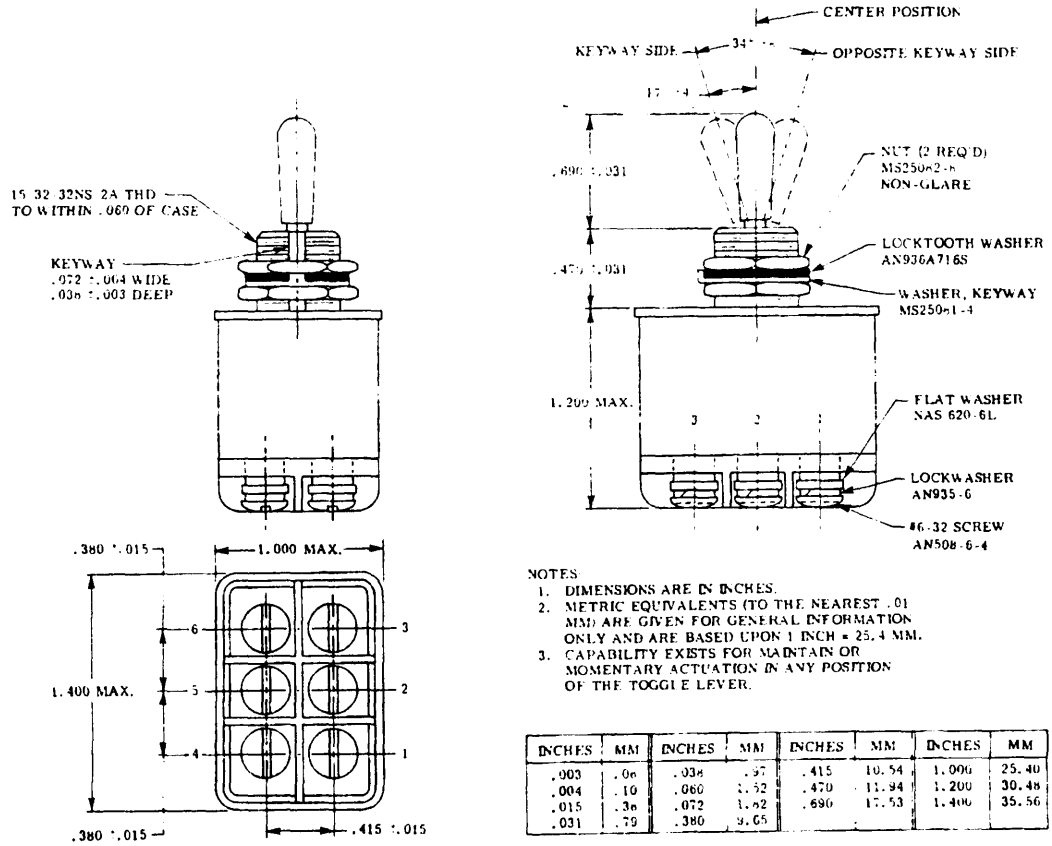
5 March 1970

* SUPERSEDING
MIL-S-81551/2(AS)
7 March 1968
(See below)

MILITARY SPECIFICATION SHEET

SWITCH, TOGGLE, DOUBLE POLE, HERMETICALLY SEALED

The complete requirements for procuring the switches described herein shall consist of this document and the issue in effect of specification MIL-S-81551.



NOTES
1. DIMENSIONS ARE IN INCHES.
2. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
3. CAPABILITY EXISTS FOR MAINTAIN OR MOMENTARY ACTUATION IN ANY POSITION OF THE TOGGLE LEVER.

INCHES	MM	INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.038	.97	.415	10.54	1.000	25.40
.004	.10	.060	1.52	.470	11.94	1.200	30.48
.015	.38	.072	1.82	.690	17.53	1.400	35.56
.031	.79	.380	9.65				

TABLE I

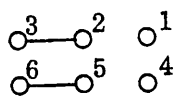
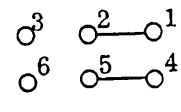
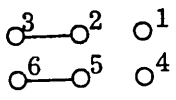
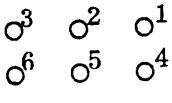
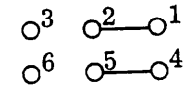
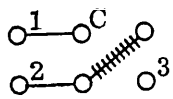
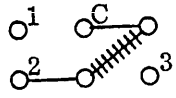
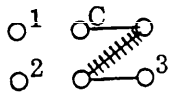
Part Number	Circuit With Toggle Lever In			Circuit Diagram	Current Capacity Amperes 28 Volts D. C.			Current Capacity Amperes 115 Volts, 60 & 400 CY. A. C.		
	Opposite Keying Side	Center	Keying Side		Resistive	Inductive	Lamp Load	Resistive	Inductive	Lamp Load
M81551/1-412	On	Off	On	II	10	5	2	10	5	2
M81551/1-462	On	None	Mom. On	I	10	5	2	10	5	2
M81551/1-552	Mom. On	On	Mom. On	III	10	5	2	10	5	2
M81551/1-512	On	Off	Mom. On	II	10	5	2	10	5	2

* (The 7 March 1968 issue was inadvertently numbered slash 2.

FSC 5930

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TABLE II

Circuit Diagram Number	Switch Function	Circuit With Toggle Lever In		
		Opposite Keyway Side	Center	Keyway Side
I	Two position 2PDT		No position	
II	Three position 2PDT			
III	Three position 1P3T selector			

EXTERNAL INSULATED JUMPER, PERMANENTLY ATTACHED

REQUIREMENTS:

Dimensions and configuration	See figure
Enclosure design	Hermetic (Class 5, MIL-S-8805)
Temperature characteristics	-65° F to 275° F non operating -65° F to 160° F operating
Shock type	MIL-E-5272, Procedure IV (50G)
Vibration grade	MIL-E-5272, Procedure XII curve A (10G)
Weight	0.30 pound maximum
Material	All external metal parts shall be corrosion resistant

OPERATING CHARACTERISTICS:

1. Insulation resistance	MIL-STD-202, Method 302 1,000 megohms, 500 VDC
2. Dielectric withstanding voltage	MIL-STD-202, Method 301
A) Atmosphere pressure	1,000 VRMS, max. leakage .5 MA
B) At 80,000 feet	400 VRMS, max. leakage .5 MA
3. Seal test	1×10^{-8} CC HE/SEC per atmosphere maximum leak rate

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OPERATING CHARACTERISTICS: (Continued)

4. Mechanical endurance	50,000 cycles
5. Electrical endurance	25,000 cycles at all current ratings
6. Electrical ratings	See Table I
7. Short circuit capability	Capable of close in on 600 amperes calibrated circuit 5 consecutive times

Custodian:
Navy - AS

Preparing activity:
Navy - AS
Project No. 5930-N279-1

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